MATHS MANTHAN
• TUITION CLASSES •
e-mail: contact@mathsmanthan.co.in | Helpline: +91-8171546558

Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – WEEKLY TEST 02
Chapter 10: Circles

Time: 45 Minutes
MM: 20

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided where indicated.
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer (VSA) question of 2 marks.
4. **Section C** comprises 3 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 1 Case Study based question of 4 marks with internal subparts.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. A quadrilateral $ABCD$ circumscribes a circle. If $AB = 6$ cm, $BC = 7$ cm, and $CD = 4$ cm, then the length of side AD is: [1]

(A) 3 cm (B) 4 cm (C) 5 cm (D) 6 cm

Q2. The length of the tangent drawn from a point A at a distance of 10 cm from the centre of a circle of radius 6 cm is: [1]

(A) 7 cm (B) 8 cm (C) $4\sqrt{2}$ cm (D) $2\sqrt{34}$ cm

Q3. If a chord AB subtends an angle of $60°$ at the centre of a circle, then the angle between the tangents at A and B is: [1]

(A) $60°$ (B) $90°$ (C) $120°$ (D) $150°$

Q4. At one end of a diameter AB of a circle of radius 5 cm, tangent XAY is drawn. The length of chord CD parallel to XY at distance 8 cm from A is: [1]

(A) 4 cm (B) 6 cm (C) 8 cm (D) 5 cm

Q5. **Assertion (A):** If a chord of the larger of two concentric circles touches the smaller circle, then it is bisected at the point of contact.
Reason (R): The perpendicular drawn from the centre of a circle to a chord bisects the chord. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (1 × 2 = 2 Marks)

Q6. In a circle with centre O, tangents PA and PB are drawn from an external point P. If $\angle APB = 70°$, find the measure of $\angle OAB$. [2]

SECTION C: Short Answer Type (3 × 3 = 9 Marks)

Q7. Prove that the parallelogram circumscribing a circle is a rhombus. [3]

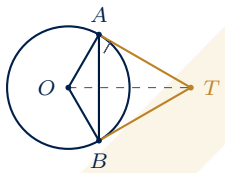
Q8. In two concentric circles, a chord AB of the larger circle touches the smaller circle at P. Prove that P bisects AB. If radii are 10 cm and 6 cm, find the length of chord AB. [3]

Q9. A right triangle ABC right-angled at B has $AB = 8$ cm and $BC = 6$ cm. A circle with centre O is inscribed in $\triangle ABC$. Find the inradius r and the area of the incircle. [3]

SECTION D: Case-Based / Applied Competency Question (1 × 4 = 4 Marks)

Q10. Bicycle Sprocket and Chain Drive Geometry

A bicycle transmission system features a circular chain sprocket of radius 6 cm with centre O. The upper and lower chain runs act as tangents TA and TB from an external derailleur guide T. The distance from the sprocket centre to the guide is $OT = 10$ cm.



(i) Determine the length of the tangent chain segment TA. [1]

(ii) Find the value of $\sin(\angle OTA)$ in the right $\triangle OAT$. [1]

(iii) Calculate the length of the chord of contact AB between the tangent points, OR determine the perimeter of quadrilateral $OATB$. [2]

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